DETERMINING SEASHELL DIVERSITY THROUGH OBJECT RECOGNITION USING THE YOLOV8 MODEL

A Thesis
Presented to the Faculty of
College of Information and Communications Technology
STI West Negros University

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

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December 2024

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ACKNOWLEDGEMENT

We want to express our heartfelt gratitude to our thesis coordinator, Ms. Orvilla V. Balangue, and our thesis adviser, Mr. Lynol I. Ibarra, for their unwavering guidance, support, and patience throughout this research journey. Their expertise and belief in our potential have been invaluable, and their encouragement has played a significant role in shaping the outcome of this work. We are truly thankful for their dedication and mentorship.

We also thank our thesis review panelists, Mr. Feleciano L. Santiago Jr., Ms. Danica S. Duazo, and Mr. Gerical Noel Travilla, for their insightful comments, constructive criticism, and encouragement. Their thoughtful feedback has enhanced the quality of this thesis, and we truly appreciate their time and effort in reviewing our work.

We would like to express our heartfelt gratitude to our parents for their unconditional love, support, and sacrifices. Their unwavering belief in us has been a constant source of motivation. We also thank our friends and loved ones for their continuous support, encouragement, and understanding. Their friendship has made this journey not only more rewarding but also more enjoyable.

Finally, we want to acknowledge our inspiration, Mr. El Jireh P. Bibanco, whose work has sparked our passion for this field. His dedication to research has been a guiding light throughout this journey. We are also grateful for the valuable input of our senior, Mr. Christian Moncera, whose expertise and guidance have significantly enriched this research. Their contributions played a key role in enhancing the quality of this work.

ABSTRACT

Title of Research:	**Determining Seashell Diversity through Object Recognition Using YOLOV8 Model**
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Degree:	**Bachelor of Science in Computer Science**
Date of Completion:	**January 2025**
Keywords:	**Counting Seashell, Occluded Objects, Object recognition, Seashell classifying, YOLOv8-n**

Determining seashell abundance is essential for assessing marine biodiversity and ecosystem health, as seashells are reliable bioindicators of water quality, pollution, and environmental change. This study presents an automated seashell classification and counting model using the YOLOv8 nano variant, focusing on three species: Mud Clam, Green Mussel, and Manila Clam. The model, trained on a carefully curated and annotated image dataset, achieved an impressive accuracy of 97.37%, demonstrating high precision, recall, and F1 scores. The methodology involved a rigorous process of data collection and augmentation techniques, followed by iterative training using a combination of optimization algorithms, including SGD, Adam, and AdamW. These efforts contributed to the model's strong performance and robust generalization. While the model excelled in controlled environments, challenges such as occlusion and environmental noise limited its performance in real-world scenarios. Aligned with the United Nations Sustainable Development Goals (SDGs) on Life Below Water (Goal 14) and Climate Action (Goal 13), the study underscores the potential of AI-driven tools in marine conservation. It advocates for enhancing annotation standards, integrating the model into mobile platforms, and extending its application to diverse ecological settings to support broader environmental monitoring. This work represents a significant contribution to AI for the environment, offering a scalable solution for seashell monitoring and laying a foundation for future advancements in marine ecosystem assessment.

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CHAPTER I
INTRODUCTION

Among the different seashells, which vary in shapes, sizes, colors, and classes, bivalves and gastropods benefit the marine environment and biodiversity. Seashells act as filter feeders by breaking down and consuming organic matter, which is beneficial in maintaining our bodies of water quality (Vaughn & Hoellien, 2018). According to Cho et al. (2021). and Curren et al. (2023), bivalves and gastropods were used as bioindicators to monitor microplastic pollution and identify microplastic contamination levels and characteristics. Therefore, it is significant to acknowledge the prominent role of the bivalve and gastropod species.

A key focus in seashell studies is quantifying to measure abundance, which provides valuable insights for marine research investigation. For instance, in 2023, Popović et al. reported that ten million tons of waste seashells were often discarded in the sea or sent to landfills. This study emphasizes the importance of quantifying seashells to address the growing issue of waste seashells. On that note, researchers can gain insights into potential threats to the marine environment by understanding the abundance (Setiawan et al., 2013; Jumawan et al., 2015; Zhang Long et al., 2010). Therefore, this study can be used to develop AI models using YOLOv8, an architecture utilizing computer vision techniques and machine learning algorithms to identify and localize objects in images and videos with remarkable speed and accuracy, suitable for classifying seashells accurately, thereby reinforcing counting tasks and providing a more efficient and reliable method for quantifying seashell abundance (Ultralytics, 2023). YOLOv8's key innovations include enhanced feature extraction, superior multi-scale object detection, and developer-friendly

enhancements, such as its unified Python package and CLI, which streamline model training and deployment (Yaseen, 2024).

Background of the Problem

Studying the abundance of seashells can be a bioindicator in marine environments and diversity. It highlights the valuable insights gained from determining the abundance of seashells to establish the relationship between seashell diversity and the marine environment, as demonstrated by researchers from Indonesia who explored the distribution of scallop seashell abundance using the Morisita distribution index, reporting that scallop seashells clustered or spread out unevenly in the waters, found in groups, thus proving uneven conditions in the marine environment (Setiawan et al., 2013). In addition, researchers from Florida explored the local abundance and distribution of a selected large gastropod species. The gastropods were grouped into three families, and the abundance of each species within these families was measured through sampling. One among the three families of gastropods was determined to be the most abundant, while the others were in low densities, revealing a disproportionate distribution of each species, indicating an equivalent imbalance in the marine environment (Stephenson et al., 2013). In the Philippines, researchers have been involved in similar undertakings. For instance, Jumawan et al. (2015) explored the abundance and distribution of clam shells along the coastline of four sites in Lanao del Norte. Clam shells were counted, measured, and identified using the transect-quadrant method. Among these four sites, clam shells were found to be the most abundant in some areas, while they have low densities in others. The study reported that anthropogenic causes, such as exploitation, and environmental parameters, such as salinity levels and sediment quality, are suggested to be the

leading causes of the variation, and concluded that the statistical result of low seashell population densities in other sites aligns with a similar study by Pacyao and Marquez (2022) in the intertidal zone of Davao Occidental Province, which determined the seashell composition, abundance, and population density. They found that there were eight different families of seashells. The study revealed that among these eight families, only one bivalve family had the highest relative abundance and population density compared to other families with low abundance. For abundance with the value, the impact of environmental parameter factors that affect species composition and abundance of seashells was highlighted. By integrating advanced machine learning techniques, this research aims to explore further models of seashell classification and counting approaches and their implications for marine environmental conditions.

Overview of the Current State of the Technology

Many studies have explored AI technologies in facilitating marine life and environment monitoring to assess marine health and life diversity and enhance research capabilities. For instance, Rasmussen et al. (2017) developed vision-based counting wild scallops using an Autonomous Underwater Vehicle (AUV) with an implemented AI model of You Only Look Once (YOLOv2) Architecture, which was then used for classifying wild scallops and reinforced counting tasks to measure population health. Additionally, using AI to facilitate counting tasks for marine species helps produce better data accuracy and eliminates human labor. However, despite these efforts, Nguyen et al. (2021) identified challenges in seashell identification and counting. For instance, scholars developed seashell recognition and counting tools for fisheries officers in New Zealand. However, the datasets were collected at a local fish market, limiting recognition among these seashells: cockle,

tuatua, and mussel. With the premise of the current limitations of seashell classification and counting as highlighted by these studies, the researchers were motivated to build models for enhanced seashell classification to reinforce counting tasks, focusing on specific seashell species utilizing the latest YOLO iteration (YOLOv8) for computer vision and neural network model to perform automated counting accurately.

Objectives of the Study

The researcher's primary objective in this study was to develop an object recognition model using a YOLOv8 nano variant architecture to accurately classify and quantify seashell species, achieving an overall performance measure above 85% (Terven et al., 2023). This model aims to provide insights into marine ecosystem health and biodiversity. Specifically, this study aimed to:

1. Determine the performance and comparison of YOLOv8 nano variant Optimizer Algorithms in seashell classification and counting tasks and
2. Apply and use the seashell classification and counting model through a mobile application.

Assumption of the Study

This section outlines assumptions that serve as a foundation for interpreting the collected and analyzed data. Recognizing and understanding these assumptions is crucial for evaluating the findings.

1. The model relies on sufficiently varied datasets for training, ensuring its ability to count numerous seashells. Having adequate training data is essential for the model to handle classified objects and counting meticulously (Hadi Khorshidi,

2021). This implies that the dataset covers the randomly scattered seashells in the finite area, with a sand background and lighting conditions. If the training data is limited, the model will not achieve better seashell counting accuracy, leading to a less effective and unreliable tool for marine studies.

2. Accurate labeling of training images is crucial to avoid errors, leading the model to learn inaccurate patterns. It is fundamental for a supervised machine learning model as it learns from these labeled examples. Labeling datasets with errors produces incorrect learning patterns, leading to inadequate performance and reliability (Shubham Gupta, 2023). This emphasizes the essence of meticulously labeling data, ensuring accuracy, and labeling seashells in the datasets, enhancing the ability to recognize for better counting accuracy.
3. The model's effectiveness relies on similar pattern characteristics and data points but contains many different values—for instance, a dataset of distributed seashells in the coastal areas. The model struggles to learn meaningful features if datasets are highly diverse and lack common patterns. This emphasizes the importance of a dataset representative of real-world scenarios (Akshit Mehra, 2024).

Significance of the Study

The significance of the study primarily benefits government and non-government marine conservation organizations, the malacological research community, and academic researchers (Charmaine Distor, 2021). Developing and implementing a model designed to enhance and reinforce current processes represents a significant advancement in seashell counting technology, offering improved insights into its performance and applications.

Government & non-government marine conservative organization. Proposals for enhancing and restoring shellfish aquaculture on the West Coast aim to restore the declining population of mollusk that lives in shallow, nearshore waters, implying that organizations strive to foster conservation for marine biology and the environment. Developing advanced technologies, as the objective of this study, enables more effective conservation strategies; AI applications facilitating marine biology must be implemented as a tool for better resource management and enforcement of environmental regulations.

Research Academe. Scholars and researchers in universities and colleges conduct investigations to push the boundaries of knowledge in their respective fields. For instance, FMCS-Freshwater Mollusk Conservation is a research community dedicated to conserving and advocating freshwater mollusks. The community stated the importance by which scientists have described over 700 species of snails and 300 species of freshwater mussels. While species are associated with every type of aquatic habitat, the most remarkable diversity is found in flowing streams. Furthermore, freshwater mollusks are a fascinating part of our natural heritage and have global significance.

Society & Community. The study can benefit local communities, especially those living near coastal areas. Using the YOLOv8 model to identify and count different types of seashells helps scientists and locals understand the health of the marine environment. The disappearance or imbalance of certain seashell species may indicate environmental impacts from pollution, overfishing, or climate change, and analyzing metadata on class, time, and quantity helps identify patterns in species abundance

over time (Sheikh Idrees et al., 2019). The study contributes to preserving marine life and supporting sustainable development for people dependent on these ecosystems.

Future Researchers. Encouraged to explore more innovative technologies to address marine conservation challenges effectively. Investigations into biodiversity and the ecological importance of mollusks should continue, ensuring their preservation and sustainability (Allison Miller et al., 2023). Building on these findings to enhance marine conservation practices and promote community engagement in sustainability efforts is vital.

Scope and Limitations of the Study

This study contributed to global efforts that aligned with the United Nations Sustainable Development Goals (SDGs), mainly "Life Below Water" and "Climate Action" (United Nations, 2012). The general purpose of the study was to develop an automated system for seashell classification and counting using the YOLOv8 model, focusing on specific seashell species such as Manila Clam, Green Mussel, and Mud Clam.

A self-made dataset of scattered and distributed seashell images was collected to evaluate the effectiveness and performance of the model, along with the development of a mobile-based system as part of the study to implement the built model for automated seashell classification and counting tasks. This system was a data-driven tool, particularly beneficial for researchers investigating seashell abundance and diversity. The target beneficiaries of the study included marine scientists, environmental researchers, and coastal resource managers (Ellen Ditria et al., 2022).

The study spanned the period of dataset collection, model training, and system development.

The study encountered several challenges beyond the researchers' control. Factors such as species procurement, financial constraints, and quality of materials used influenced the environmental settings and potentially impacted the model's accuracy. Additionally, the placement and arrangement of scattered seashells posed a challenge, as their variability could affect the system's performance in real-world scenarios. While the system was designed to adapt to diverse conditions and incorporate a broader range of seashell species, future enhancements will require expanding its applicability to a wider range of automated seashell classification and counting tasks (Andrew Leon et al., 2010).

Review of Related Literature/Studies/Systems

This chapter covers a comprehensive review of literature related to seashell counting that significantly correlates to AI facilitating marine biology. The literature is divided into the following areas: Image processing and AI techniques, AI in marine biology, Seashell counting and Biodiversity, and AI in Marine Studies in the Philippines. Additionally, it provides challenges and limitations that facilitate a comprehensive understanding of the study.

Image Processing and AI Techniques. Deep learning has transformed image processing, enabling digital images to be manipulated using computer algorithms for tasks such as classification, segmentation, and recognition (Kundu, 2022) while also encompassing techniques like enhancement, restoration, and analysis to achieve diverse results (Pandey, 2023), unlocking valuable information for applications in

fields like security and surveillance. Image analysis techniques are becoming increasingly significant in ecological studies and crucial for population monitoring and habitat evaluation (Kulkarni, 2023).

Deep learning techniques significantly enhance image processing. One prominent example is the convolutional neural network (CNN), a general-purpose deep learning architecture for various image-related tasks like classification, segmentation, and recognition (Mandal, 2024). While CNNs can also be used for object detection, specialized deep learning architectures like Faster Region-Convolutional Neural Network (Faster R-CNN), You Only Look Once (YOLO), and Mask Region-Convolutional Neural Network (Mask R-CNN) leverage CNNs for feature extraction, which is crucial for efficiently and accurately identifying objects in images (Rizzoli, 2021).

Several studies have applied AI-driven image analysis with object detection algorithms for counting tasks in ecological studies. For instance, Liu et al. (2023) proposed a novel, dense multi-scale feature aggregation network to address the challenge of significant variations in fish size for counting fish in deep-sea aquaculture. Moreover, Guirado et al.(2019) proposed a Convolutional Neural Network (CNN) for automated whale detection and counting in satellite and aerial images based on open data to address the lack of knowledge about whale populations. Furthermore, Rassmussen et al. developed an Autonomous Underwater Vehicle (AUV) System based on deep learning YOLOv2 for vision-based counting of wild scallops to measure population health.

This study highlights the potential use of AI-driven image processing, particularly object detection algorithms, to count marine organisms in images accurately. This

technology can significantly enhance marine research capabilities and advanced marine life monitoring. Implementing seashell counting using object detection algorithms offers key benefits in marine studies, like automatic and accurate population assessments, which can play a crucial role in developing more effective conservation strategies for marine ecosystems (Ellen Ditria et al., 2022).

Artificial Intelligence in Marine Biology. Dogan et al. (2024) highlighted the emergence of AI in marine biology, such as oceanographic forecasting, marine species monitoring, coral reef assessment, and AI-enabled marine robotics. AI facilitating marine biology covers key benefits such as enhanced data analysis, predictive monitoring, and providing better resource management. Moreover, several studies have explored the significant potential of AI for marine biology and environmental research. For instance, counting the number of fish using the traditional sampling method is time-consuming and laborious and may put pressure on the fish. Therefore, Zhang et al. (2020) proposed an automatic fish-counting method based on a hybrid neural network combining multi-column convolutional neural network (MCNN) and dilated convolutional neural network (DCNN) to achieve a real-time and better accuracy of counting fish in far offshore mariculture.

Furthermore, Chen et al. (2023) found that plankton are widely distributed in the aquatic environment and indicate water quality. Monitoring variation in plankton is an efficient approach to forewarning environmental risk. However, counting at a microscopic level is time-consuming and laborious. Therefore, scholars proposed an automated video tracking system based on deep learning to continuously monitor living plankton abundance in aquatic environments. This study highlights the positive potential of AI in marine biology, leading to technological advancements (Tuomas Eerola et al., 2024). Implementing seashell counting through AI offers key benefits,

such as advanced monitoring and improved research capabilities, ultimately leading to more effective marine conservation strategies.

Seashell Counting and Biodiversity. Seashell abundance is estimated using sampling techniques, such as random sampling, to count seashells and analyze their composition, population density, and distribution patterns, ensuring unbiased results and enabling statistical estimation of total abundance (Pacyao & Marquez, 2022; Lasari & Harahap, 2022; Thomas, 2023). Furthermore, Jumawan et al. (2015) used the transect-quadrat method to assess clam shell abundance and distribution, highlighting the importance of sampling techniques for accurately estimating marine life abundance and gaining valuable ecological insights (Study Mind, 2023).

Seashells offer valuable insights into marine diversity and ecology, serving as indicators of habitat type and water quality (Agro, 2021; Curen et al., 2023; Bouzahouane et al., 2024), with studies highlighting gastropods as water quality indicators based on their division into nine families, two of which are strongly linked to salinity levels and clean freshwater (Alhejoj et al., 2017), and bivalves as bioindicators of microplastic pollution, which directly and indirectly affect their physiology, habitat, food sources, and organic pollutant persistence (Khanjani et al., 2023).

This study emphasizes the significant relationship between seashell indicator species and their abundance. Studying seashell abundance provides additional insights into marine diversity. This suggests that seashells play a crucial role in coastal ecosystems, and their life cycles are integral to the food web system.

AI in Marine Studies in the Philippines. Researchers in the Philippines have conducted extensive marine studies, including marine life monitoring and population

assessment, with Dejeto et al. (2016) evaluating planktivorous fish composition and biomass alongside phytoplankton composition and density in the outer part of San Pedro Bay, Leyte, Philippines, where they observed significant variation in fish biomass, particularly in Palo, a local area with the highest fish abundance during the rainy season, suggesting that food source availability significantly influences fish abundance and highlighting the importance of life monitoring as an indicator of food availability.

Implementing AI in facilitating marine studies improves research capabilities and advanced data monitoring, especially in utilizing AI for Philippine researchers. For instance, Wang et al. (2023) discussed that the semantic segmentation of marine images is more straightforward to describe seafloor scenes and track marine organisms. However, creating human-annotated datasets for image segmentation requires time and effort. Therefore, they proposed a semi-supervised learning approach that combines the mean-teacher model and U-net architecture for segmenting seafloor images and monitoring in Pujada Bay, Philippines. The proposed method performs segmentation categories, including corals, sea, urchins, sea stars, and seagrass, emphasizing the engagement of exploration seafloor monitoring. Furthermore, Montalbo et al. (2019) proposed the VGG16 Deep Convolutional Neural Network (DCNN) for fish species classification in the Philippines for aquaculture and marine ecology protection.

This study highlights the significant impact of AI facilitating marine studies in the Philippines. This technology helps marine researchers conduct further studies to improve research capabilities and monitor marine life, which is crucial in technological advancements.

Challenges and Limitations

Occlusion of objects. Seashells on the seabed are rarely neatly separated. They can be piled on top of each other or obscured by other marine life. This occlusion makes it difficult for an AI to differentiate individual shells and accurately count them in an image (Chilukuri et al., 2022).

Potential Biased AI Counting. The performance of any AI model is highly dependent on the data it is trained on. If the training data for seashell counting primarily consists of images of shells in specific orientations, lighting conditions, or species, the AI may struggle to identify shells that appear differently. This bias can lead to undercounting certain species of shells or missing them entirely (Silberg & Manyika, 2019; D'Alessandro et al., 2023).

Environmental Factors. Variations of objects within the frame from controlled environment settings can significantly impact the AI's ability to detect seashells. Factors like lighting variations and background clutter (rocks, sand) can affect how seashells appear in an image. Due to these environmental factors, the AI might misidentify other objects as shells or miss them altogether (Study Mind, 2023).

Specific Limitations to the Philippines Context. The Philippines is a hotspot for marine biodiversity, with many seashell species. However, there may be limited availability of training data images that accurately represent the diversity of these species. If the AI is trained on data from a different geographical location, it might not accurately identify and count the unique seashells in the Philippines (Agduma et al., 2023).

Definition of Terms

This section provides definitions of key terms used in this research ensure clarity and consistency.

Adam. Adaptive Moment Estimation: Optimizes by combining momentum and adaptive learning rates to improve efficiency and convergence.

AdamW. Adaptive Moment Estimation with Weight decay combines Adam's adaptive learning rates with weight decay to enhance generalization and prevent overfitting.

Ground-truth. The true label or value of an input is used as a reference for model evaluation and data understanding.

Hyperparameter. A configuration setting for a machine learning model that is set before the learning process begins, such as learning rate, batch size, or number of layers. Hyperparameters control the training process and can significantly affect model performance.

Instance. A single, specific occurrence of a class (or object) that YOLOv8 is trained to detect, whether during training, validation, or inference. Each instance corresponds to one object in an image or dataset, with its associated bounding box and class label.

Object Noise. An object in an image that does not belong to any defined class but possesses characteristics that the model misinterprets as belonging to a classified category.**Hyperparameter Tuning.** The process of systematically adjusting and selecting the best hyperparameters (such as learning rate, batch size, etc.) to optimize the performance of a machine learning model. This process involves iterating over different combinations to achieve the best configuration.

Occlusion. Happens when an object in an image is partially hidden or blocked by something else, like when a car is partly covered by a tree or a person is standing

behind a wall. This can make it harder for object detection models, such as YOLOv8, to recognize and accurately locate the object. By training these models on images with examples of occluded objects, we can help them become better at identifying and understanding objects in real-world situations where things are rarely perfectly visible

Optimizer Algorithm. A mathematical method is used to update the weights and biases of a model during training to minimize the loss function. Examples include SGD, Adam, and AdamW. These algorithms operate on data subsets, making them faster and better suited for large datasets while also helping to escape local minima.

SGD. An optimization algorithm that updates parameters iteratively using random

YOLO. A real-time object detection model that processes entire images in a single pass to detect and classify objects efficiently.

CHAPTER II
METHODOLOGY

This chapter provides context for the research method, detailing the datasets used and the strategy applied. As Penny Mathieson et al. (2018) emphasize, a well-structured research plan and careful selection process lay the foundation for reliable and meaningful results. By following procedures and addressing potential limitations and biases, researchers ensure the credibility and integrity of findings in seashell counting.

Research Design

This section outlined the strategy for conducting an experimental analysis and data-gathering procedures for the datasets (Srivast et al., 2020). This approach established a proper setting to uncover potential insights, including the model's performance and the variables present in the datasets. This comprehensive research design enabled the researchers to thoroughly describe and interpret the results of the model's performance and the influence of relevant variables within the chosen datasets, ensuring a well-rounded understanding of the findings (Ultralytics, 2023).

Datasets

The researchers established data-gathering procedures by placing a box frame, measured up to a 36-square-inch area, to capture the entire view of a single seashell within the frame. This setup simplifies data collection and facilitates feeding the data into a model for counting tasks. The YOLOv8 object-detection nano variant, specifically designed for object detection, was deemed suitable for seashell counting tasks. These counting tasks focused on classifying a limited variety of seashell species: Mud Clam, Manila Clam, and Green Mussel. The results of the seashell counting

methods were labeled to indicate the total number of seashells counted. This approach ensured accurate data annotation and consistency across the dataset, as illustrated in Table 1. for the non-augmented and augmented datasets.

Table 1. Non-Augmented Instance Dataset split

Data type	Training	Validation	Testing	Total
Manila Clam	8106	1450	627	10,183
Mud Clam	2438	446	178	3,062
Green Mussel	5607	1087	480	7,174
Total	16,151	2,983	1,285	20,419

Table 2. Augmented Instance Dataset split

Data type	Training	Validation	Testing	Total
Manila Clam	49814	1450	627	51,891
Mud Clam	49980	446	178	50,604
Green Mussel	49881	1087	480	51,448
Total	149,675	2,983	1,285	153,943

Data Gathering Instrument and Setup Process

The instruments and equipment for the data gathering setup would be a box frame, measured 6 x 6 square inches with orange tarpaulin as an image background, two light stands, and a DSLR Nikon-d3200 camera, with 18-120mm camera lens on a camera

tripod stand to avoid image distortion for capturing the image. The camera stand height is 1ft to capture the whole image view within the box frame. This data gathering setup will be established in a controlled environment, preferably inside the 7x7x5 feet tent, to avoid natural disruption like sunlight. Figure 1 shows the layout used to gather data.

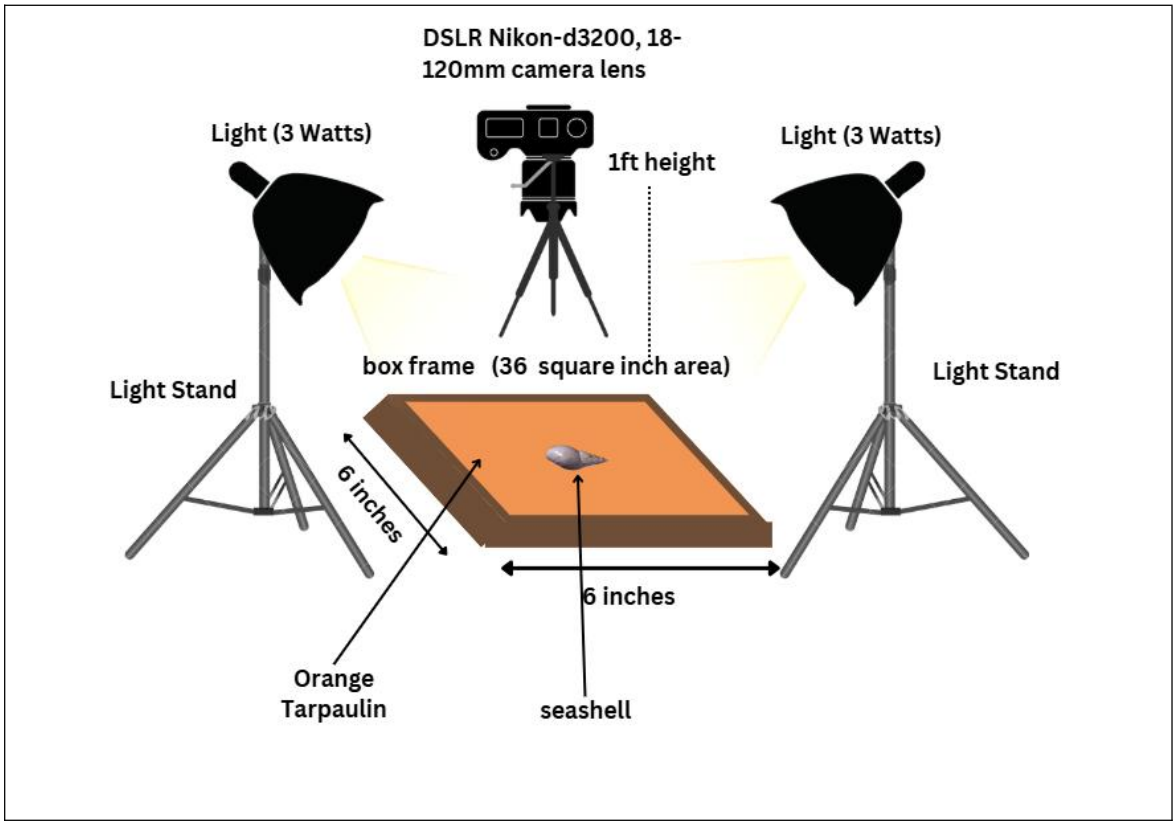


Figure 1. Data gathering setup

Measures

This section discusses the measurements to be used to evaluate the model. The variables within the formulas, derived from the testing phase of the procedure, are discussed later in this chapter. Formulas of measurements and standards to be followed are also known and were used for object detection applications, respectively. Intersection over Union (IoU) serves as a basis for evaluating the performance of the object detection model for seashell counting. Vedoveli (2023) highlights IoU as a

pivotal metric in computer vision, particularly for tasks such as object detection and segmentation, emphasizing its critical role in assessing the quality and accuracy of models. Specifically, IoU quantifies the model's ability to localize objects precisely within images, offering a robust evaluation of its detection capabilities.

In object occlusion, IoU evaluates how effectively the model handles overlapping or partially hidden objects, such as seashells obscured by one another in captured images. IoU metric quantifies the degree of overlap between the predicted bounding boxes and the ground truth bounding boxes, particularly emphasizing scenarios where objects are partially visible. By calculating the ratio of the intersection area to the union area of these two bounding boxes, IoU provides a numerical value reflecting the model's ability to accurately localize and separate individual objects, even under challenging occlusion conditions (Ultralytics, 2023).

$$\text{IoU} = \frac{Area\ of Overlap}{Area\ of\ Union} = \frac{|B_p \bigcap B_{gt}|}{|B_p \bigcup B_{gt}|}$$

In addition to measuring Intersection over Union (IoU), other essential evaluation metrics, such as Precision, Recall, and F1-Score, serve as foundational tools for assessing the performance of object detection models (Ultralytics, 2023). These metrics provide a comprehensive understanding of the model's ability to detect and classify objects accurately, complementing IoU evaluation. The calculations for these metrics rely on the concepts of True Positives (TP), False Positives (FP), and False Negatives (FN). True Positives refer to correctly detected objects where the predicted bounding box aligns with the ground truth. False Positives indicate instances where the model incorrectly identifies an object where none exists, and False Negatives represent missed detections where the model fails to identify a present object. Table 3 presents the confusion matrix classification, which compares the

model's predictions against the ground-truth values. It categorizes the predictions into True Positives, False Positives, and False Negatives for each class, providing insight into how well the model correctly identifies or misclassifies objects.

The researchers used this matrix to help evaluate the performance of a model by providing various metrics such as accuracy, precision, recall, and F1 score. Accuracy is calculated by dividing the total number of correct predictions (TP and TN) by the total number of instances in the dataset. A higher accuracy score indicates that the model can more accurately categorize instances from both positive and negative groups. It is an overall measure of the model's performance in correctly classifying cases.

$$\text{Accuracy} = \frac{TP}{TP + FP + FN}$$

Precision measures the proportion of how well the model distinguishes accurate optimistic predictions from predictions made. In other words, it measures how many of the instances predicted as positive are positive. (Amer, 2022).

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures the proportion of accurate optimistic predictions from all actual positive instances in the dataset. It measures how many positive instances were correctly identified by the model (Amer, 2022; Ultralytics, 2023).

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1 Score provides a balanced measure of the model's performance, considering false positives and negatives. This metric is handy when there is an imbalance between positive and negative classes in the datasets (Amer, 2022; Ultralytics, 2023).

$$F1 = 2 \times \frac{precision \times recall}{precision + recall}$$

In addition, Mean Average Precision (mAP) plays a crucial role in evaluating the overall performance of object detection models by combining precision and recall across different thresholds and classes. This involves averaging the precision scores calculated at various recall levels for each class in the dataset. Essentially, mAP examines the model's ability to consistently and accurately detect and classify objects across a range of conditions and object categories (Molugaram & Rao, 2017). Mean Average Precision or mAP's comprehensive evaluation integrates precision-recall curves to provide a robust performance measure (Ultralytics, 2023).

$$\text{Mean Average Precision} = \frac{1}{n} \sum_{k=1}^{k=n} AP_k$$

Procedures

Training. YOLOv8 was chosen for its lightweight efficiency and strong community support, enabling flexible deployment. Pretrained models provided a solid foundation, followed by iterative hyperparameter tuning across four optimization strategies—SGD, Adam, and AdamW—to identify the best-performing configurations. Each optimizer represented its top-performing model during training.

The training phase spanned 100 epochs without early stopping, allowing a complete exploration of underfitting, optimal fitting, and overfitting. This thorough observation guided fine-tuning to improve performance and generalization.

Augmentation. Augmentation focused on ensuring the model could adapt to real-world conditions, addressing critical factors like camera distance, image quality, and

lighting. YOLOv8's built-in augmentation was complemented by task-specific adjustments, enhancing the model's usability and accuracy (Alhassan Mumuni et al., 2022).

Epoch Iteration. Training over 100 epochs offered a deep understanding of learning dynamics across different stages. This provided critical insights into how the model transitions between underfitting, optimal fitting, and overfitting, informing adjustments to improve learning stability and outcomes (Ultralytics, 2023; Aditya Kumar, 2024).

Optimizer Selection. The training process evaluates three primary optimizers chosen for everyday object detection tasks, each offering a unique approach to optimizing learning rates and model performance. These optimizers were selected to align with the study's objectives:

- Stochastic Gradient Descent (SGD): SGD is a baseline optimizer that offers simplicity and reliability for foundational deep learning tasks (Handbook of Hydroinformatics, 2023).
- Adam (Adaptive Moment Estimation): Enhances convergence efficiency by combining momentum with adaptive learning rates, making it effective across diverse tasks (Machine Learning for Biomedical Applications, 2024).
- AdamW: A variation of Adam that features weight decay regularization to improve generalization and mitigate overfitting (Fernando Dijkinga, 2024).

Hyperparameter Tuning. Metrics like IoU, Precision, Recall, F1-Score, Accuracy, and mAP-guided model selection. These balanced evaluations revealed trade-offs and highlighted the most reliable configurations. Avoiding early stopping allowed a complete assessment of training dynamics, deepening understanding of

hyperparameters' impact on generalization (Justus Akinlolu, 2024).

Avoiding early stopping ensures a detailed understanding of training dynamics, empowering researchers to identify how each hyperparameter and optimizer affects generalization and fitting states. This deliberate approach fosters a deeper understanding of the trade-offs required to balance key metrics, enhancing the reliability and adaptability of the model.

Testing. The trained model's predictions, typically assessed with a standard 50% confidence threshold (Deval Shah, 2023), are evaluated with a more challenging 70% threshold, where predictions meeting or exceeding this level are classified as true positives and their bounding boxes are compared to the ground truth to calculate the Intersection over Union (IoU), while predictions falling below 70% overlap with the ground truth are classified as false positives, ensuring a systematic evaluation and revealing opportunities for model improvement (Dihao Ai et al., 2023).

Confusion Matrix for Model Evaluation. Metrics like Precision, Recall, and F1-Score complemented IoU, offering a well-rounded view of detection accuracy. The confusion matrix categorized predictions into True Positives, False Positives, and False Negatives, highlighting class-specific strengths and weaknesses.

Table 3. Confusion Matrix Classification

	Class A	Class B	Class C
True Positives	Predicted A when Actual is A	Predicted B when Actual is B	Predicted C when Actual is C
False Positives	Predicted B when Actual is A Predicted C when Actual is A	Predicted A when Actual is B Predicted C when Actual is B	Predicted A when Actual is C Predicted B when Actual is C
False Negatives	Actual A, Predicted B, C, or missed	Actual B, Predicted A, or C	Actual C, Predicted A or B

Best Optimizer Algorithm Model. It is identified through a thorough evaluation process that considers the used measurements, with the overall effectiveness determined by averaging these measurements.

Measurement by Class of Models. The best optimizer is identified through averaged performance across evaluation metrics. Performance analysis by class highlighted strengths and weaknesses, revealing areas where the model excels or requires refinement.

Framework Experimentation. In addition, the many actions and development methods employed by the researchers throughout the study, guided by a framework, are discussed in this section. Based on Ramasubramanian and Singh, Plan, Explore, Build, and Evaluate make up the four stages of the PEBE project management system. From early planning to project evaluation, this framework is intended to offer an organized and iterative approach to project management, as shown in Figure 2.

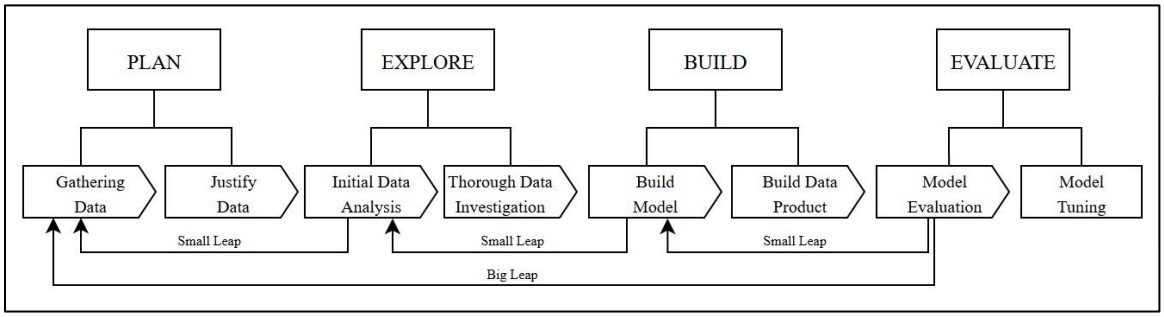


Figure 2. PEBE Framework

Plan. During the initial planning stage, the project researchers defined specific objectives, goals, and problems they aimed to address, which involved quantifying seashells in a controlled environment. They outlined the expected outcomes and identified the necessary resources for the project, such as target datasets, computational power, and expertise in machine learning.

Explore. Yiding Jiang (2023) explains that the exploration phase involved investigating existing image processing and machine learning studies, focusing on methods and algorithms for identifying multi-object scenarios while examining approaches for collecting and annotating image datasets to train and evaluate the machine learning model.

The researchers created and individually annotated the datasets, adhering to guidelines from Michael Abramov (2024) to ensure at least 30% object visibility and a maximum IoU threshold of 40% during annotation, with any inconsistencies or biases, such as insufficient occurrences of occlusion, addressed and re-evaluated in the planning phase to enhance dataset quality.

Build. The build stage marked the implementation of the proposed model, which integrated the YOLOv8 object detection nano variant's pre-trained model for seashell counting, with the selection of this variant driven by its computation speed and the

minimal difference in performance between variants, which is significant in the context of the study's use case (Gaudenz Boesch, 2023). The system architecture included object locating, object labeling, and counting, and following the principles outlined by Ramasubramanian and Singh, researchers created and rigorously tested multiple algorithms, optimized hyperparameters, and seamlessly integrated various system components.

Evaluate. The researchers will thoroughly examine the effectiveness of their machine-learning model in detecting instances of individual seashells. It will thoroughly assess a carefully selected dataset, using necessary evaluation measures such as IoU, precision, recall, F1 score, and overall average (Ultralytics, 2023).

To verify that the model was not simply treading familiar ground, we benchmarked it against leading methodologies in the area, gaining a clear grasp of its efficacy compared to the current state-of-the-art, and guided by the perceptive ideas established by Ramasubramanian and Singh (2018), we performed a critical analysis to examine the work's strengths and shortcomings, extending the evaluation beyond quantitative measurements to explore qualitative features and gain insight into the model's complex benefits and limits, identifying specific areas for development and refinement, while acknowledging that the complexity of real-world situations presented a challenge, as simply altering settings based on the held-out dataset would be insufficient (Abdul-Razak Gamani, 2024).

The iterative feedback loop constantly tuned the model based on its performance in real-world scenarios, accounting for data intricacies and ensuring the model's output aligned with the complexity of the actual use case (Ramasubramanian & Singh, 2018), and this process of evaluation, refinement, and re-evaluation was critical in increasing

the model's robustness and performance, paving the road for its successful use in real-world circumstances.

By proceeding beyond a simple performance assessment and adopting a diverse evaluation process, the efficacy of the seashell counting model was ensured while laying the groundwork for its ongoing improvement and adaptation in the face of real-world challenges (Andrew Leon et al., 2010). This commitment to extensive examination highlighted responsible and effective AI development characteristics.

CHAPTER III
RESULTS AND DISCUSSIONS

This chapter explores the results and insights gained from evaluating four distinct object detection models optimized with Adam, AdamW, and SGD algorithms across 12 model iterations. The primary evaluation metric, mAP@0.50-0.95, measured performance. Initially, the Adam optimizer delivered the highest mAP score (84.25% at epoch 67), but after a deeper analysis, the SGD optimizer performed best overall, achieving an average score of 95.60%. The best-performing SGD model also excelled in other metrics, including an IoU of 88.26%, precision of 96.88%, recall of 98.99%, accuracy of 95.93%, and an F1-score of 97.92%. One key takeaway was that training with non-augmented data slightly outperformed models trained with augmented datasets by 0.39%, demonstrating that under certain conditions, less data manipulation can still produce highly efficient models. A mismatch between annotation criteria and model detection capabilities affected objects with less than 30% visibility, impacting evaluation outcomes. This finding highlights the need to align annotation methods with assessment criteria. Model performance was examined across different classifications, including Manila Clam, Mud Clam, and Green Mussel. These insights emphasize the importance of refining both data preparation techniques and evaluation methods to improve object detection accuracy in practical applications. The following sections will dive deeper into these findings and their broader implications in object detection research.

Hyperparameter Tuning and Best Model Selection

Four models are produced in this phase, each representing a different optimizer algorithm, with 12 models built. The goal is to select the best model based on the hyperparameter tuning process. The performance of these models is evaluated using mAP@0.50-0.95, precision, and recall, which averages the mean Average Precision (mAP) over multiple Intersection over Union (IoU) thresholds ranging from 0.50 to 0.95 (in increments of 0.05). This provides a more comprehensive evaluation compared to mAP@0.50 alone, as it considers a broader range of IoU thresholds, capturing model performance over varying levels of overlap between predicted and ground truth boxes (Ultralytics, 2023).

Hyperparameter Tuning of Adam Optimizer Algorithm using Augmented Dataset. The best-performing model emerged during the first iteration, achieving a mAP@0.50-0.95 score of 84.25% at the 67th epoch. Then, the second iteration at the 61st epoch scored 84.10%, the fourth iteration at the 95th epoch with 84.06%, and the third iteration at the 58th epoch with 83.85%. Figure 3 highlights these scores, comparing the iterations and emphasizing the best-performing epoch based on mAP@0.50-0.95.

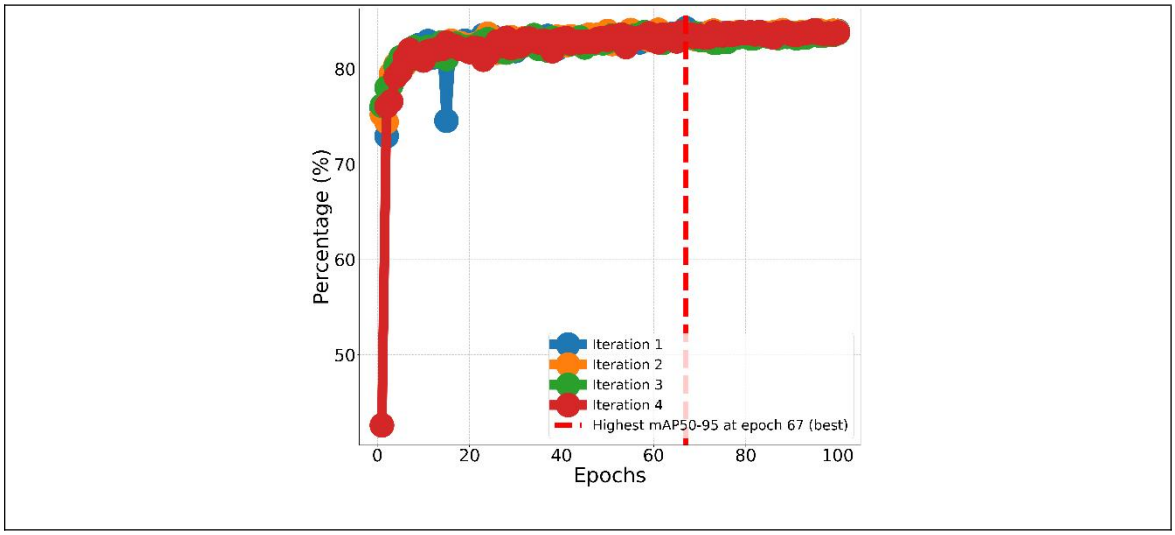


Figure 3. Adam Optimizer Algorithm Hyperparameter Tuning

Model Training of AdamW Optimizer Algorithm using Augmented Dataset. The best-performing model was achieved during the third iteration at the 73rd epoch, with an mAP score of 84.05%. This was followed by the first iteration at the 40th epoch with a score of 83.94%, the fourth iteration at the 49th epoch with a score of 83.75%, and the second iteration at the 33rd epoch with a score of 83.68%. Figure 4 highlights these scores, comparing the iterations and emphasizing the best-performing epoch based on mAP@0.50-0.95.

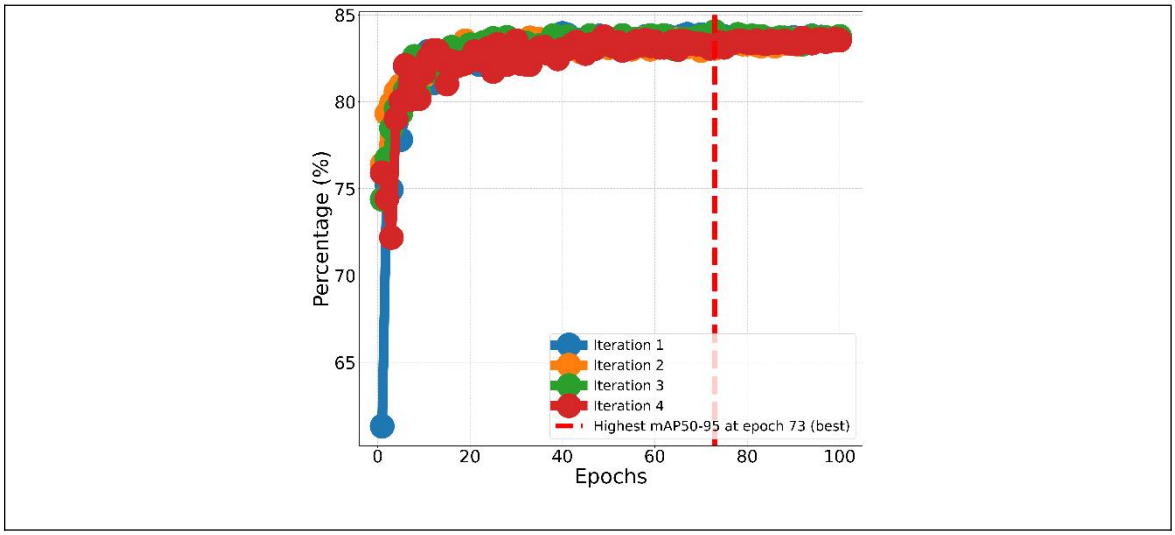


Figure 4. AdamW Optimizer Algorithm Hyperparameter Tuning

Model Training of SGD Optimizer Algorithm using Augmented Dataset. The best-performing model emerged during the fourth iteration, achieving an mAP@0.50-0.95 score of 84.00% at the 95th epoch. This was followed by the second iteration at the 100th epoch with a score of 83.91%, the third iteration at the 94th epoch with 83.80%, and the first iteration at the 85th epoch with 83.47%. Figure 5 highlights these scores, comparing the iterations and emphasizing the best-performing epoch based on mAP@0.50-0.95.

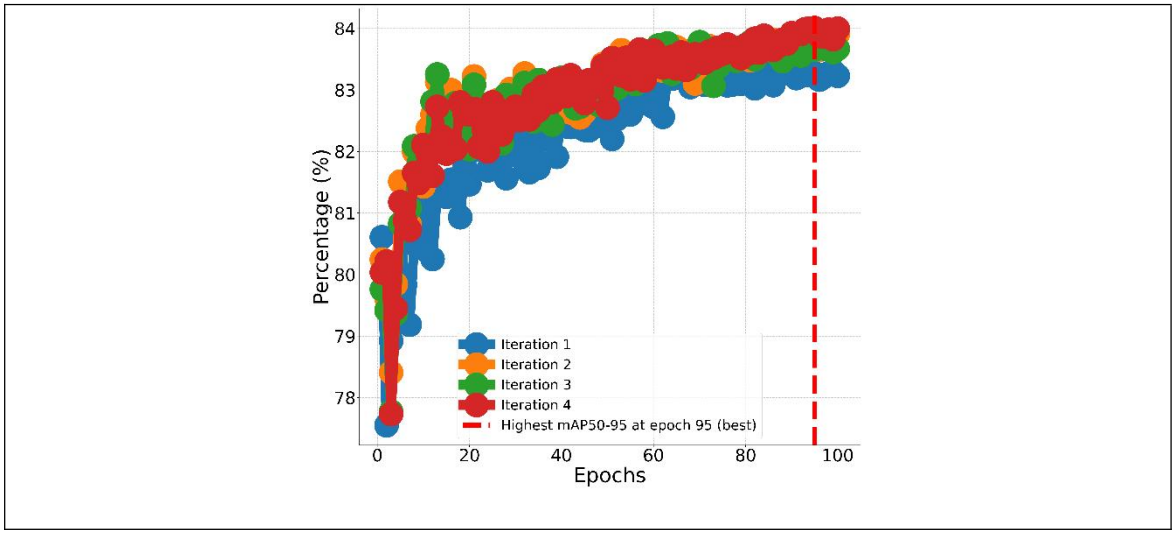


Figure 5. SGD Optimizer Algorithm Hyperparameter Tuning

The analysis highlights the outstanding performance of the Adam Optimizer Algorithm, whose top-performing model achieved the highest mAP@0.50-0.95 score of 84.25% during the 67th epoch in the first iteration of training. This score surpassed the AdamW Optimizer Algorithm, whose best model recorded an mAP of 84.05% at the 73rd epoch in the third iteration, and the SGD Optimizer Algorithm, which secured an mAP of 84.00% at the 95th epoch in its fourth iteration. These results underscore the effectiveness of the Adam Optimizer Algorithm in producing consistently superior models during hyperparameter tuning with the augmented

dataset, as evidenced by its ability to edge out the AdamW and SGD optimizer algorithms by consistently achieving the highest mAP@0.50-0.95 scores.

Evaluation of Top-Performing Models from Optimizer Algorithms

Comprehensive testing was conducted to further evaluate the top-performing models each optimizer algorithm produced, extending beyond the mAP@0.50-0.95 metric, which, while serving as a robust aggregate score derived from various measurements, necessitates deeper exploration into individual performance metrics to better understand each optimizer algorithm's behavior, as each algorithm employs distinct logic and strategies during training, potentially influencing aspects such as precision, recall, or training efficiency, and by analyzing these metrics separately, we achieve a more holistic evaluation of the optimizer algorithms' capabilities, providing insights into their strengths and limitations that mAP@0.50-0.95 alone cannot, following Xiang Lei's (2024) approach, which helps researchers gain deeper insights into models' performance through visualizations.

Measure Comparison. Among the different optimizers explored—SGD, AdamW, and Adam—SGD emerged as the most effective, achieving a striking average of 95.60%, with strong metrics including 88.26% IoU, 96.88% precision, 98.99% recall, 95.93% accuracy, and 97.92% F1-score, demonstrating superior balance between detection accuracy and localization precision, with precision reflecting its strong ability to classify positive samples, recall indicating its efficiency in identifying all relevant objects, and its IoU measure further affirming the model's capability to localize occluded objects to near 100% accuracy and precision, while accuracy and F1-score reinforced its consistent performance, highlighting its robustness in object classification and overall detection tasks, and Figure 6 summarizes these results,

comparing optimizers based on IoU, Precision, Recall, F1-Score, and overall average measures, following the general rule of thumb to focus on mAP for comparing models' performance, as noted by Neurocomputing (2023).

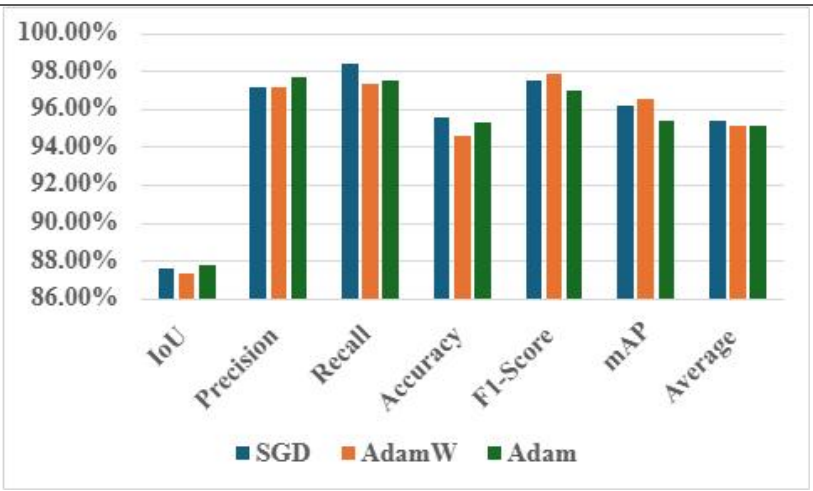


Figure 6. Optimizer Performance Comparison

In complement to Neurocomputing, 2023's "widely recognized metric used in the field of information retrieval and performance evaluation," the study showed the top-performing optimizer algorithm through an average measure comparison. This algorithm was used in training a new model with a non-augmented dataset, which provided a critical baseline for evaluating the optimizer's ability to generalize and adapt without data augmentation. Training on non-augmented data clarified the algorithm's raw performance in learning meaningful patterns from the original dataset. The comparison showed that data augmentation did not significantly improve the model's accuracy and robustness, as the model trained with the SGD optimizer on the non-augmented dataset outperformed models trained with augmented datasets by an average of 0.39%. This result demonstrated the optimizer's exceptional adaptability and efficiency, confirming its reliability even without augmented datasets. Figure 7 visually compares these results, highlighting performance differences across the models.

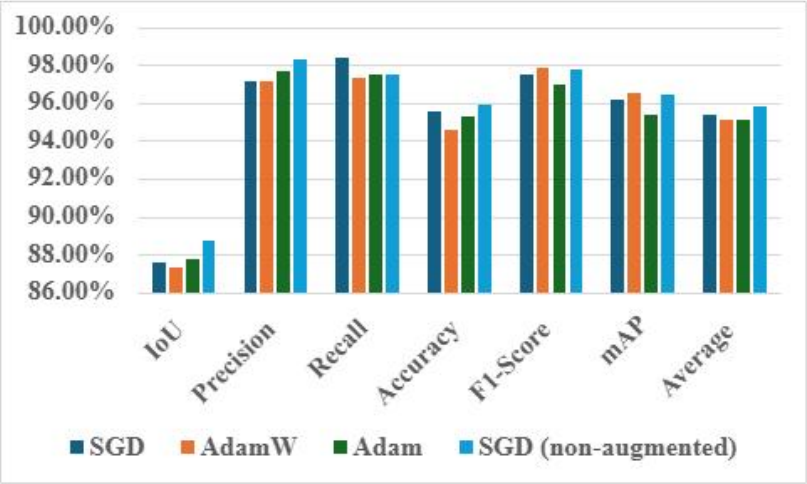


Figure 7. Optimizer Performance Comparison with SGD Optimizer Algorithm using Non-Augmented Dataset

Comparison of Ground-truth and Predicted labels

Following Deval Shah, 2023's guidelines for choosing the optimal threshold, findings have emerged, revealing that a deeper analysis of the results showed the model predicted objects with less than 30% visibility due to being trained in a controlled environment, which caused it to mistakenly identify "noise objects" as legitimate classes, classifying both a "noise object" as an actual object and a "true object" with greater than 30% visibility, with Figure 8 illustrating an example where the model incorrectly predicted a ground-truth non-labeled object.

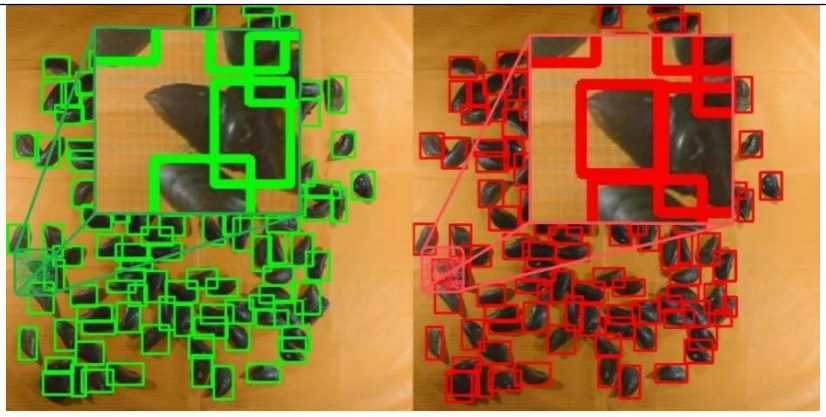


Figure 8. Comparison Between Ground-Truth Labels and Model Predictions for Unlabeled Objects

Concerning the experimentation and investigation, this may be a sign of underfitting or overfitting. This occurrence is not a sign of underfitting because the model learned meaningful patterns, especially as the dataset was collected using a medium occlusion approach. On the other hand, overfitting is also unlikely because the model demonstrated the ability to generalize beyond the training data by detecting partially visible objects, even those excluded by the strict visibility threshold imposed during annotation. This suggests that the model is not merely memorizing the training data but can handle real-world scenarios, albeit with a misalignment between its learned capabilities and the evaluation criteria (Ultralytics, 2023).

Annotation Criteria and Model Detection Misalignment

The scenario shown in Figure 8 is not unique but extends across various species and classes, resulting in similar misclassifications, which undermine the computational validity of the evaluation metrics as the system incorrectly registers these instances as False Positives, with the misalignment stemming from a conflict between the annotation criteria used in the methodology and the model's limitations in detecting objects with less than 30% visibility, a situation that Jace Cruz (2023) highlights as a potential source of errors during the phase of training a model, such as annotating.

Measure Comparison of Model Performance by Class

This section provides a detailed comparison of model performance across different optimizer algorithms by class, focusing on key evaluation metrics: Precision, Accuracy, Recall, F1 Score, and Intersection over Union (IoU). The SGD-optimized model on the augmented dataset showed strong performance, but it was outperformed in several metrics by models using the AdamW optimizer. Precision, for example, indicated that the non-augmented SGD model outperformed the augmented one,

particularly in the Manila Clam and Green Mussel classes. Regarding accuracy, recall, and F1 score, the AdamW-optimized models showed marginally higher values across all classes. Similarly, the IoU scores followed a similar pattern, with the Adam-optimized model achieving slightly better results than SGD, though the differences were minimal. The following figures (Figures 9–13) visually compare the performance of the models for each class, offering a clearer understanding of the differences in model behavior.

Precision. The SGD-optimized model on the augmented dataset achieves 96.96% precision in Manila Clam, 99.44% in Mud Clam, and 97.12% in Green Mussel. However, the model that utilizes the SGD optimizer algorithm using a non-augmented dataset outperformed it in precision by 98.19% in Manila Clam, 99.44% in Mud Clam, and 98.32% in Green Mussel. Figure 9 presents the comparison of Optimizer Algorithm models by each class.

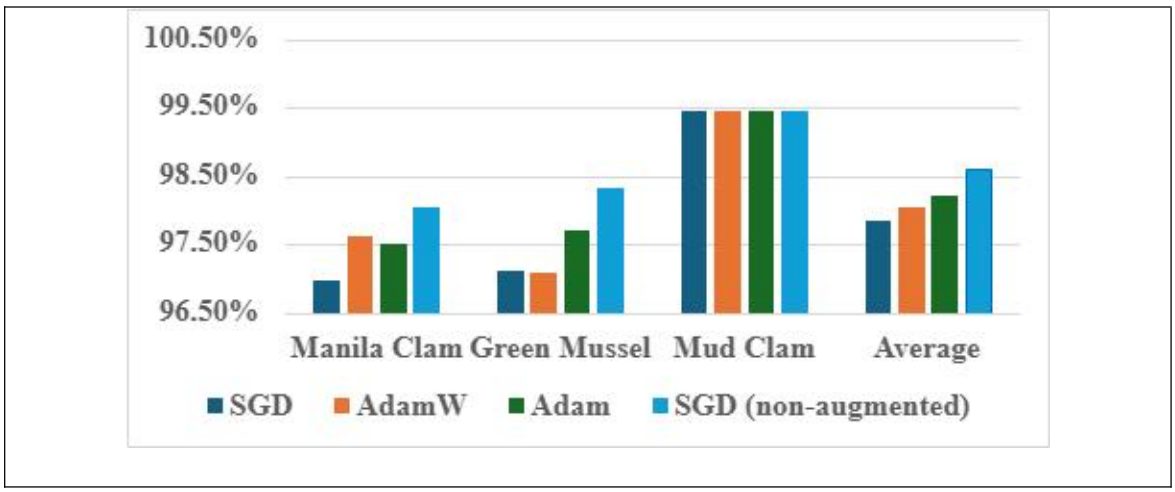


Figure 9. Precision Comparison of Model Optimizer Algorithms by Class.

Accuracy. The SGD-optimized model on the augmented dataset achieves 93.64% accuracy in Manila Clam, 99.44% in Mud Clam, and 95.55% in Green Mussel. However, the model that utilizes the AdamW optimizer algorithm outperformed it in accuracy by 95.63% in Manila Clam, 99.44% in Mud Clam, and 94.53% in Green

Mussel. Figure 10 presents the comparison of Optimizer Algorithm models by each class.

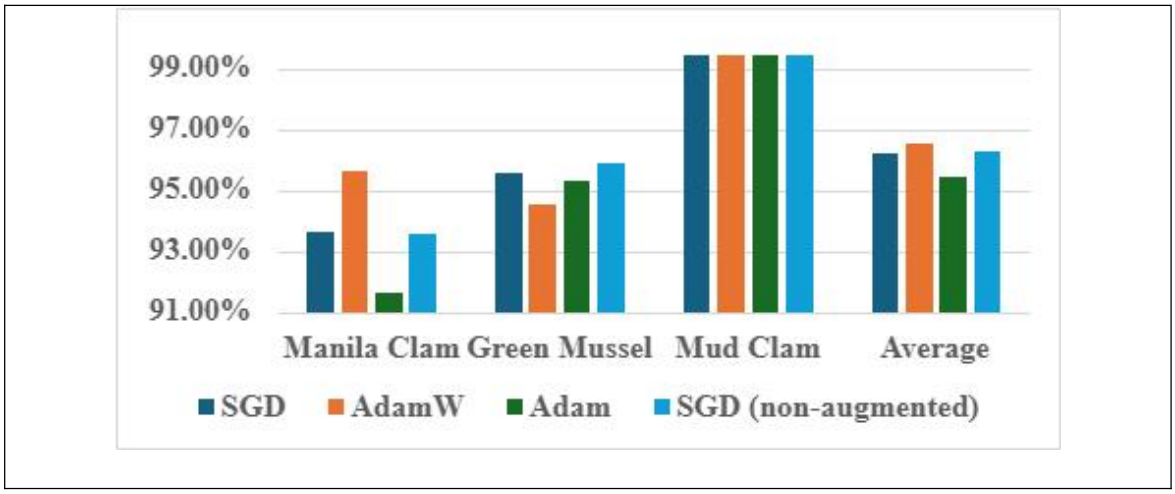


Figure 10. Accuracy Comparison of Model Optimizer Algorithm by Class

Recall. The SGD-optimized model on the augmented dataset achieves 96.49% recall in Manila Clam, 100% in Mud Clam, and 98.33% in Green Mussel. However, the model that utilizes the AdamW optimizer algorithm outperformed it in the recall by 97.93% in Manila Clam, 100% in Mud Clam, and 97.29% in Green Mussel. Figure 11 presents the comparison of Optimizer Algorithm models by each class.

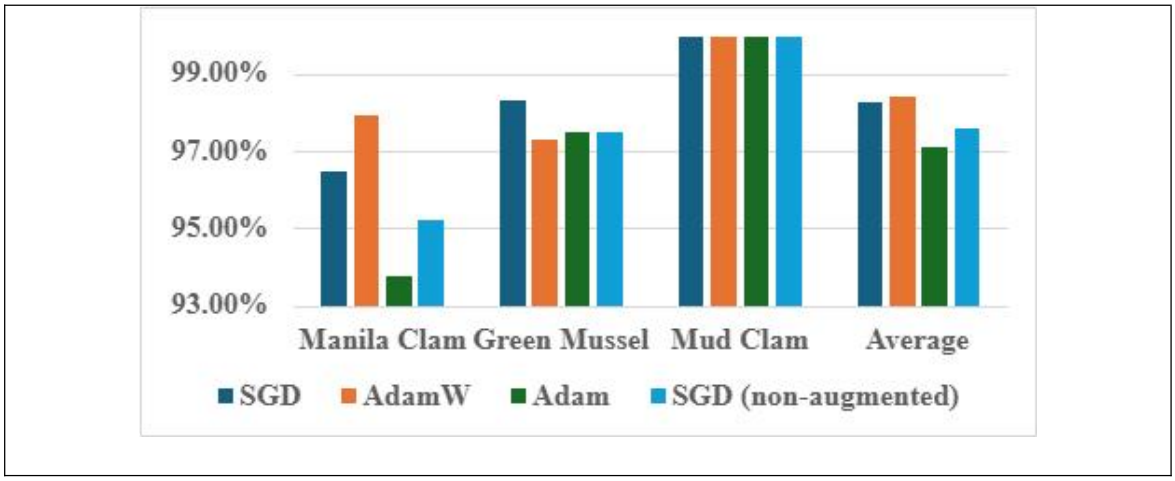


Figure 11. Recall Comparison of Model Optimizer Algorithm by Class

F1. The SGD-optimized model on the augmented dataset achieves a 96.72% f1 score in Manila Clam, 99.72% in Mud Clam, and 97.72% in Green Mussel. However, the model that utilizes the AdamW optimizer algorithm outperformed it in the recall by

97.77% in Manila Clam, 99.72% in Mud Clam, 97.19% in Green Mussel, and 0.05% in average difference. Figure 12 presents the comparison of Optimizer Algorithm models by each class.

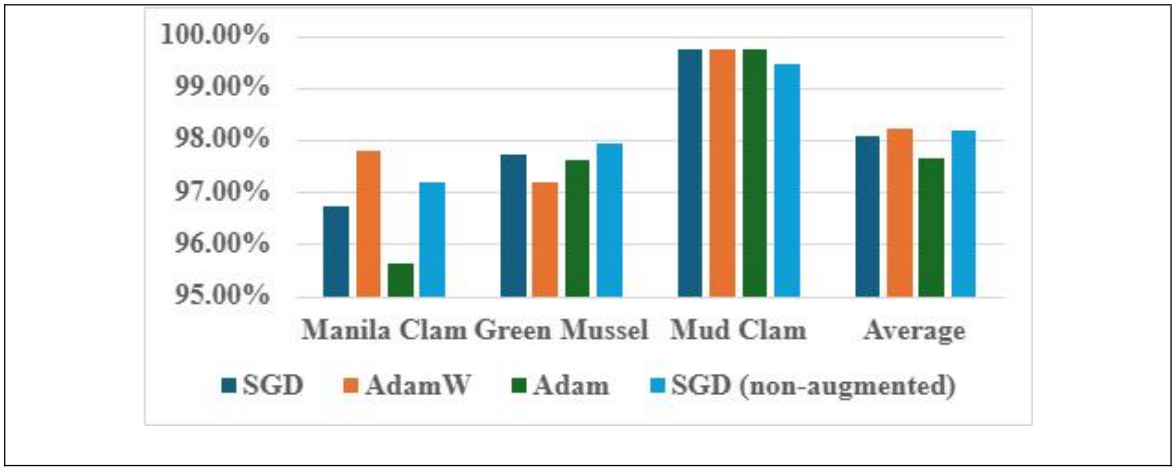


Figure 12. F1 Score Comparison of Model Optimizer Algorithm by Class

IoU. The SGD-optimized model on the augmented dataset achieves 90.45% IoU in Manila Clam, 93.33% in Mud Clam, and 88.72% in Green Mussel. However, the model that utilizes the Adam optimizer algorithm outperformed it in the recall by 90.03% in Manila Clam, 92.87% in Mud Clam, 87.76% in Green Mussel, and a 0.61% average difference. Figure 13 presents the comparison of Optimizer Algorithm models by each class.

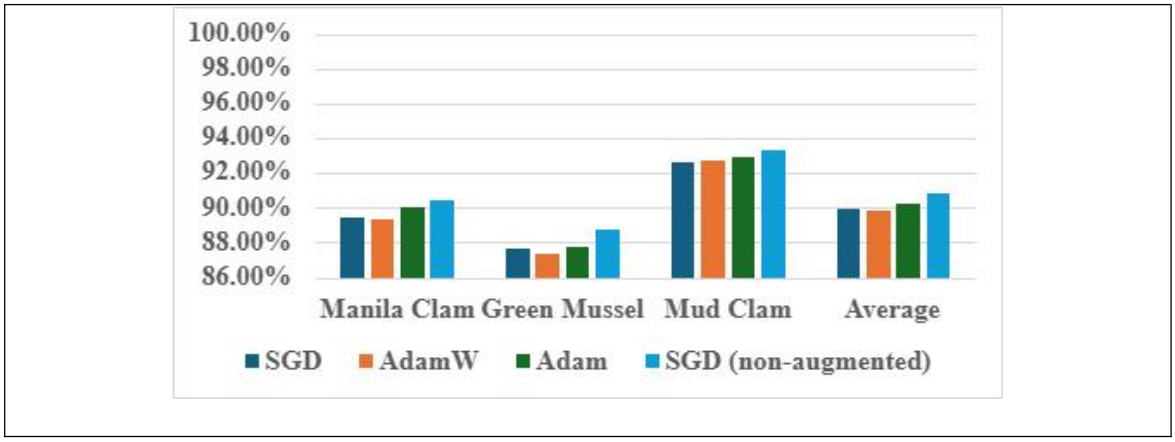


Figure 13. IoU Comparison of Model Optimizer Algorithm by Class

CHAPTER IV
CONCLUSION AND RECOMMENDATIONS

The study successfully created a machine learning model that is capable of classifying and quantifying seashell species accurately leveraged by the YOLOv8-n model. The model achieved the best iteration in 97.37% average combined with the Precision, Accuracy, Recall, and F1-score, demonstrating its prominent potential for marine biodiversity monitoring and ecological assessments.

A key insight was the potential impact of the dataset boxing annotation threshold that causes the model to produce insightful limitations during evaluation metrics. The strict 30% visibility requirement eliminates visible objects partially, limiting the model's capability to detect such cases. Adjusting the box annotation threshold to 10-15% during model testing could better align the annotation process with the model's strength, potentially producing better results.

Despite these parameters, YOLOv8-n exhibited notable robustness in terms of training performance. It produces consistency through augmented and non-augmented datasets, maintaining high accuracy even with minor human errors during testing. The model also excelled at detecting occluded objects in under a certain condition, particularly when optimized with SGD, AdamW, and Adam.

Testing in uncontrolled environment may presented challenges such detecting “object or background noises” which causes misclassification that has similar characteristics or unclassified objects. These real-world limitations underscore the need for further refinements, such as the training parameters and techniques on more diverse datasets to enhance the model’s training on a generalization

approach. Understanding the model's training dynamics proved a critical insight. Identifying key epochs' need for underfitting, proper fitting, and overfitting allowed for data-driven augmentation, optimizing performance, and offering a roadmap for future enhancements.

The YOLOv8-n model emerged as a powerful model for both classification and object detection research approaches, combining robust performance in processing due to its well-structured architecture for image processing. It excelled in identifying seashell species and holds promise for broader environmental monitoring applications. By addressing the issue like visibility thresholds and enhancing adaptability to real-world environmental settings, this work lays a foundation for advancements in object detection technologies.

Collaboration among researchers, developers, and other stakeholders is vital to aligning technological advancements with ecological goals. Future efforts challenge the YOLOv8 nano variant by training datasets in more dynamic settings and complex conditions. Such research may refine its capabilities and expand its real-world applications.

Ultimately, this study highlights the promise of YOLOv8-n while addressing the challenges of advancing object detection. Bridging gaps between model capabilities, annotation standard techniques, and environmental realities that provides a strong foundation for more innovate and efficient tools that may drive to enhance ecological and scientific progress.

Recommendations

Future Research Directions. The study suggested further research to optimize YOLO-v8-n model features, particularly in refining feature computations techniques

to enhance the model's accuracy, robustness, and occlusion event handling capabilities. This optimization can incorporate additional sensors or pre-trained models, such as image segmentation models, to improve performance across various conditions.

Expanding Model Applications. This study could expand the model's potential by exploring its ability to identify different objects or species beyond the current study's scope. Exploring experimentation of the model's capability, such as fine-tuning for diverse environments and object variations, would maximize its practical utility.

Integration into Existing Processes. Integrating the YOLOv8-n model into existing processes, such as marine ecosystem monitoring and environmental assessments, is strongly recommended. This integration could streamline the state-of-the-art use of object recognition through many applications, which may be incorporated to optimize workflows and minimize human error.

Development of Standalone Applications. Using this machine learning model could provide precise seashell identification and quantification by creating a standalone application, such as a smartphone app. In addition, this also applies to different platforms that require image processing requirement. This innovation has the potential to enhance transparency and build consumer trust in industries reliant on accurate environmental data.

Encouraging Collaborative Efforts. The study fosters collaboration among industry stakeholders, including engineers, researchers, students, and technology enthusiasts, to further refine and expand the model's applications.

Real-World Dataset Challenges. It is recommended to challenge the YOLOv8 object-detection nano variant by training datasets in uncontrolled environments, where external factors such as noise, rocks, pebbles, and changes in light can affect the model's performance. Investigating these factors would help better understand the model's capabilities and limitations in real-world scenarios.

Explore Threshold Annotation. We recommend challenging the YOLOv8 object-detection nano variant by training datasets in uncontrolled environments, where external factors such as noise, rocks, pebbles, and changes in light can affect the model's performance. Investigating these factors would help better understand the model's capabilities and limitations in real-world scenarios.

YOLOv8 Segmentation and Polygonal Annotation. Using the pre-trained YOLOv8-n segmentation model was suggested while maintaining the same methodology for dataset collection. However, the annotation process should focus on exploring polygonal annotations to define precise object boundaries, which is essential for effective segmentation. Incorporating polygonal annotations can improve the model's ability to handle overlapping or complex-shaped objects, enhancing its overall segmentation accuracy.

Data graph visualization. Implementing graph-based visualizations can help detect patterns and anomalies effectively, providing users with a comprehensive understanding of the system’s performance.

Saving captured images and previewing the pictures on the Logs page. Including previews of captured images on the Logs, page can assist users in identifying

anomalies within the model's outputs by evaluating the bounding boxes generated by the model.

CHAPTER V
SYSTEM OUTPUT

The system is an offline application designed to capture and process images efficiently. Its primary functionality includes saving the model's output with an attached date-time stamp indicating when the system recorded the output. This feature is crucial as it provides an apparent historical reference, enabling users to trace and organize results effectively over time. During the design phase of the system's output functionality, the emphasis was placed on ensuring users could obtain results without internet connectivity. This capability is critical when internet access is unavailable or unreliable, ensuring uninterrupted operations and usability.

The saved outputs are also structured to include a precise timeframe, enhancing accessibility and manageability. This structure simplifies data organization, enabling users to quickly identify and retrieve specific outputs for further analysis or reporting

.

Figure 14 illustrates the process of receiving the model's output. Users can save or retry captured images, and saved outputs are accessible through the Logs page (Figure 15), the "Saved Output Result" page (Figure 16), or the "Home" page (Figure 17). This design allows users to revisit previous results for analysis and reference, emphasizing a user-centric approach. By enabling secure storage and easy access to past outputs, the system enhances efficiency, reliability, and overall user experience in data management.

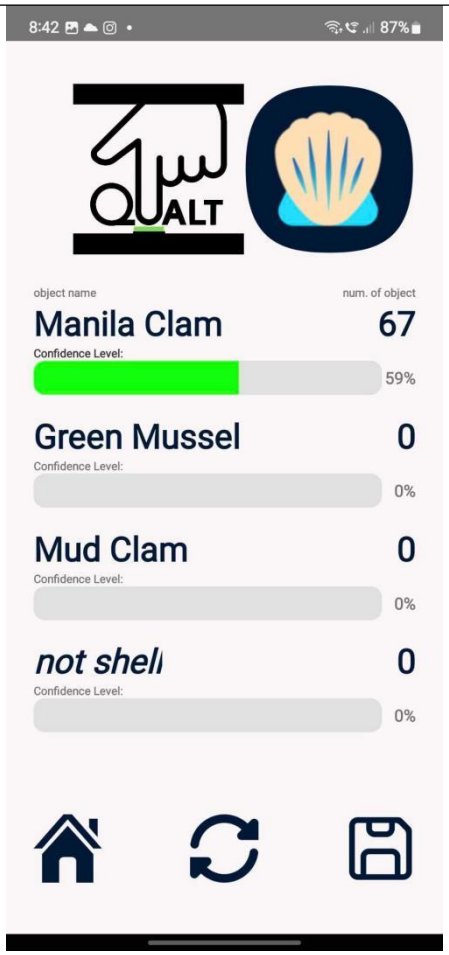


Figure 14. Model Result Page



Figure 15. Saved Output Result Page



Figure 16. Home Page



Figure 17. Log Page

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APPENDICES

Appendix A

CALENDAR OF ACTIVITIES

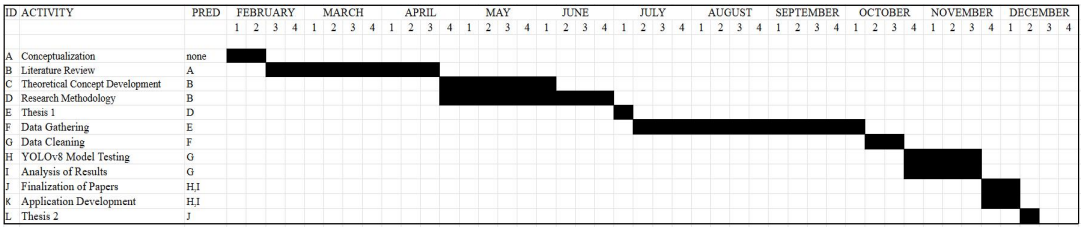


Figure 18. Gantt Chart Calendar

PERT TABLE

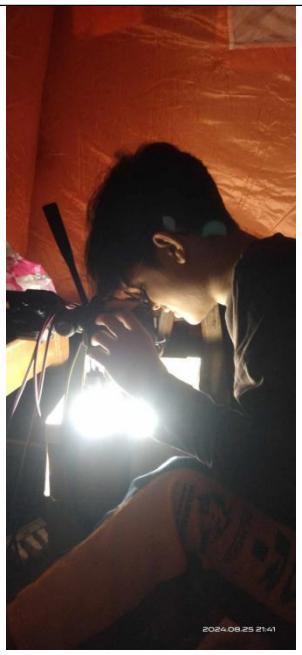
CODE	PREDECESSOR	SUCCESSOR	a	m	b	ET	ES	EF	LS	LF	SLACK
A	NONE	B	1	3	3	2	0	2	0	2	0
B	A	C, D	7	10	10	9	2	11	2	11	0
C	B	NONE	3	7	10	6	11	17	11	17	0
D	B	E	8	9	10	9	11	20	11	20	0
E	D	F	1	1	3	1	20	21	20	21	0
F	E	G	9	12	15	12	21	33	21	33	0
G	F	H, I	1	3	4	2	33	35	33	35	0
H	G	J	3	4	6	4	35	39	35	39	0
I	G	J	3	4	6	4	35	39	35	39	0
J	H, I	K	2	3	5	3	39	42	39	42	0
K	J	NONE	1	1	1	1	42	43	42	43	0

Figure 19. PERT Chart

Appendix B

DATA GATHERING PICTURES









Appendix C

ACTUAL THESIS EXPENSES

Quantity	Specifics	Approximate Cost	Actual Cost
	Equipment (3 R-socket, 3 Plug, Wire, 3 Firefly bulbs (5 Watts), 2 Headlamp, 1 Tarpaulin, 3 Service (assemble wire)		1,300
7	Seashells Specie	200	1400
2	Google Collab Pro	584.96	1169.92
3	Travel Expense	1500	4500
1	Senior Consult	700	700
3	Manuscript Draft	150	750
		Total Cost	**9,819.92**

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Lynol I. Ibarra — **Engr. Orvilla V. Balangue, MSCS**

Appendix D

USER'S MANUAL

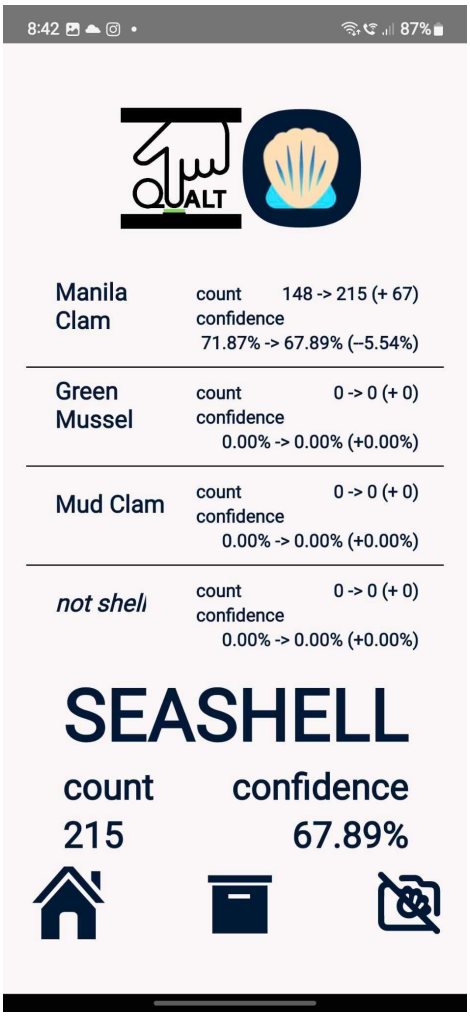
Capturing and Saving Seashell species count

(1) Capturing and Saving Seashell specie count	(2) Capturing and Saving Seashell specie count
	

(3) Capturing and Saving Seashell specie count



4) Capturing and Saving Seashell species count



Log Check, Row Delete, and Data Reset

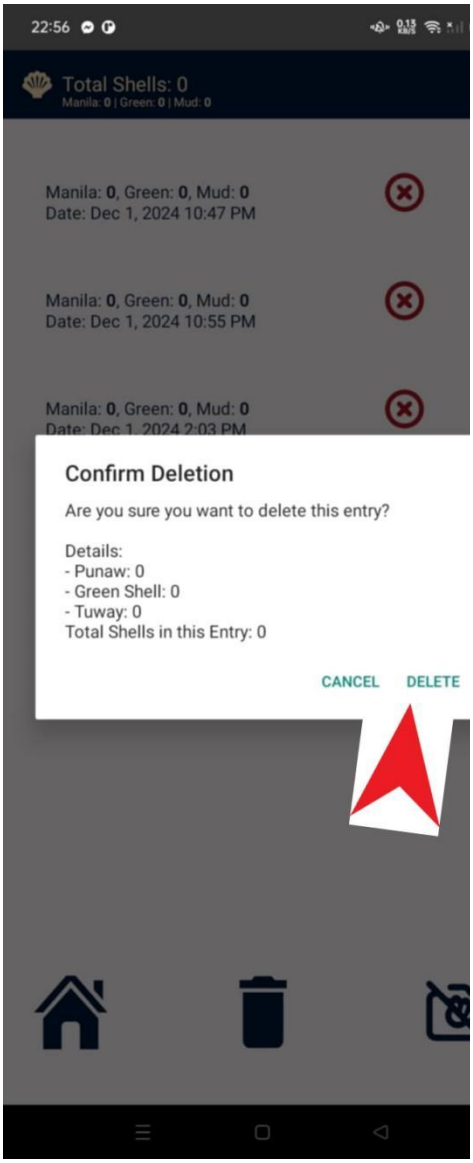
(1) Log Check, Row Delete, and Data Reset



(2) Log Check, Row Delete, and Data Reset



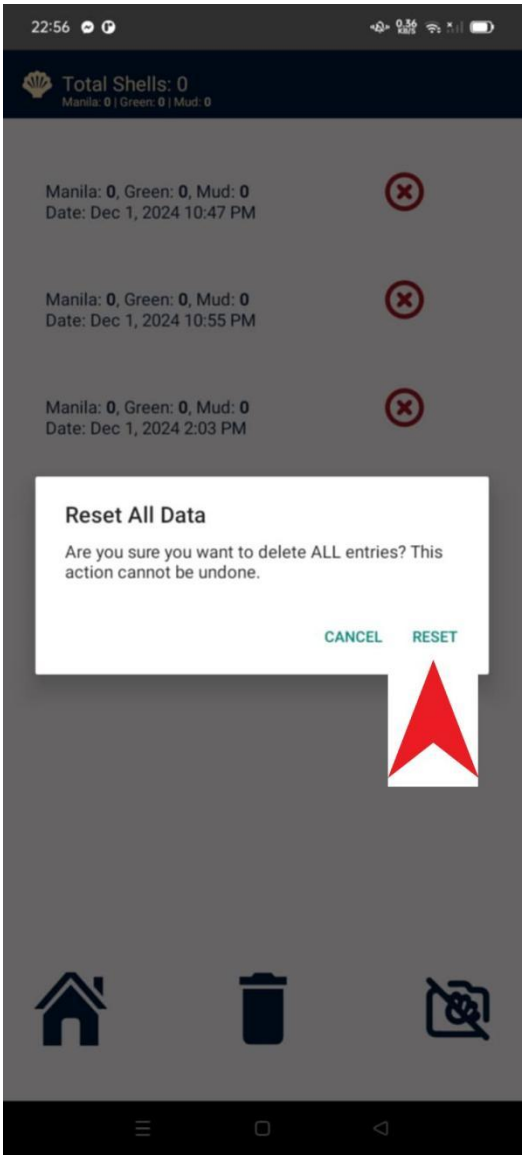
(3) Log Check, Row Delete, and Data Reset



(4) Log Check, Row Delete, and Data Reset



(5) Log Check, Row Delete, and Data Reset



(6) Log Check, Row Delete, and Data Reset



E. **ADVISER ACCEPTANCE FORM**

F. **ADVISER AND AGREEMENT**

G. **CERTIFICATE OF RESEARCH ETHICS APPROVAL**

CURRICULUM VITAE



Curriculum Vitae of

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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2025	STI West Negros University
Senior Highschool	2021	LCC-B
Junior Highschool	2019	Evangelical Christian Montessori School
Elementary	2015	LCC-B

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
September 2021 - Present	Programmer's Guild	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Software Engineering	Beginner	2022
Backend Developer	Beginner	2022
Problem-Solving Focus	Beginner	2022
Active Listening	Beginner	2022
Evaluating Profitability (CBA)	Beginner	2022

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
2023	Introduction to Python
2023	NYC



Curriculum Vitae of

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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	July 2021 - Present	STI West Negros University
High School	April 2013 -2017	St. Joseph High School Inc.
Elementary	June 2007 -2013	Tambo Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
August 2024	4th Year Chairperson	Programmer's Guild
September 2023	3rd Year Chairperson	Programmer's Guild
August 2023	Student Assistant	CICT
January 2023	Student Mentor	Programmer's Guild

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
September 2021 - Present	Programmer's Guild	Member
August 2024 - Present	4th Year Chairperson	Programmer's Guild
September 2023	3rd Year Chairperson	Programmer's Guild

SKILLS

SKILLS	Level of Competency	Date Acquired
Graphics Design	Beginner	April 2024
Unity 2D Game Development	Intermediate	March 2023
Problem-solving	Beginner	August 2019
Critical Thinker	Beginner	August 2019
Great in Communication	Novice	May 2018
UI Design	Intermediate	August 2019

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
January 2023	Concept to code: Keys to Hackathon Success Organization
March 2022	2D Game Development Using Unity



Curriculum Vitae of

NJEADH F. CAPITAN

Blk 5 Lt 4 Phase 2, Charito Heights,

Brgy. Granada, Bacolod City, Negros Occidental 6100

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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	Sept. 2023 - Present	STI West Negros University
High School	June 2015 -2019	Emiliano Lizares National High School
Elementary	June 2009 -2015	Asuncion Lizares National High School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
January 2023 - Present	Student Mentor	Programmer's Guild

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
September 2023 - Present	Programmer's Guild	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
C#	Intermediate	June 2024
Python	Beginner	June 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
Oct. 2023	Data Science vs AI Bootcamp
2023	NYC



Curriculum Vitae of

JHUNRHICK B. TARRE

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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	2019 - Present	STI West Negros University
High School	2014 -2018	Emiliano Lizares National High School
Elementary	2008 -2014	Asuncion Lizares National High School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
January 2023 - Present	Student Mentor	Programmer's Guild

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
September 2021 - Present	Programmer's Guild	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Microsoft Office 365 Literate	Intermediate	August 2019
Proficient in building Mobile/Web Application and Development	Novice	August 2019
Unity 2D Game Development	Intermediate	March 2023

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
July 2024	Online course completion in Introduction of Machine learning

August 2023	Google Devfest in Machine learning
July 2023	Concept in difference between Disinformation and Misinformation
January 2023	Concept to code: Keys to Hackathon Success Organization
November 2022	Online course completion in Introduction of ReactNative
March 2022	2D Game Development Using Unity
May 2019	Online course completion in Java Intermediate
May 2019	Online course completion in PHP
April 2019	Online course completion in Power of AJAX with JQuery
May 2018	Online course completion in Python
April 2018	Online course completion in Web Development